Skin Cancer Detection App

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Abstract:

Skin cancer is one of the most common forms of cancer worldwide, and early detection is critical for improving treatment outcomes. This project develops a deep learning-based skin lesion classification system designed to assist with early screening through image-based prediction. A convolutional neural network based on EfficientNet was trained using dermoscopic image datasets to classify lesions into clinically relevant categories. The system is designed with future mobile deployment in mind, allowing users to upload lesion images and receive rapid prediction results. The goal is not to replace dermatologists, but to provide an accessible screening tool that may support earlier medical consultation.

Problem and Motivation:

- Skin cancer cases continue to rise globally
- Early diagnosis significantly improves survival rates
- Access to dermatology specialists may be limited
- AI-based screening tools can improve accessibility
- Traditional diagnosis requires expert interpretation
- Automated systems can assist in preliminary screening

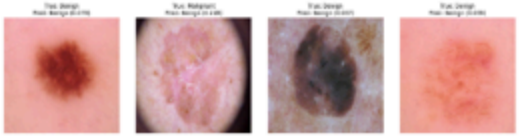
System Architecture

Dataset and Preprocessing

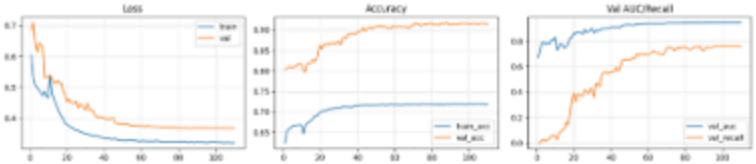
- HAM10000 skin lesion dataset used for training
- 7 lesion categories included
- Images resized to 224 × 224 pixels
- Data augmentation applied:
  - rotation
  - flipping
  - brightness variation
  - Mixup
- Dataset split into:
  - Training set
  - Validation set
  - Test set
- Class imbalance handled during preprocessing

Results:

Model performance on test dataset:



	best_val_auc	best_val_acc	chosen_thr	test_auc	test_acc	test_sens
0	0.932959	0.903869	0.420009	0.945308	0.905522	0.750853
1	0.943834	0.913247	0.564168	0.945342	0.897538	0.641638
2	0.964434	0.932005	0.462546	0.944780	0.908849	0.753788
3	0.971353	0.936694	0.544966	0.947674	0.911510	0.679181
4	0.950826	0.927315	0.387078	0.940304	0.901530	0.771331
5	0.957932	0.924883	0.465601	0.961213	0.920825	0.778157
6	0.939948	0.908451	0.435523	0.951161	0.910845	0.791809
7	0.945028	0.917840	0.506960	0.945896	0.906188	0.706485
8	0.956931	0.914319	0.331675	0.943711	0.896873	0.843003
9	0.964400	0.926056	0.426455	0.937852	0.897538	0.744027



Mobile App/Interface

The system is designed for future mobile accessibility. Users upload lesion images and receive classification output through a simplified interface.

Features:

- Image upload
- Prediction display
- Confidence score
- Fast inference time